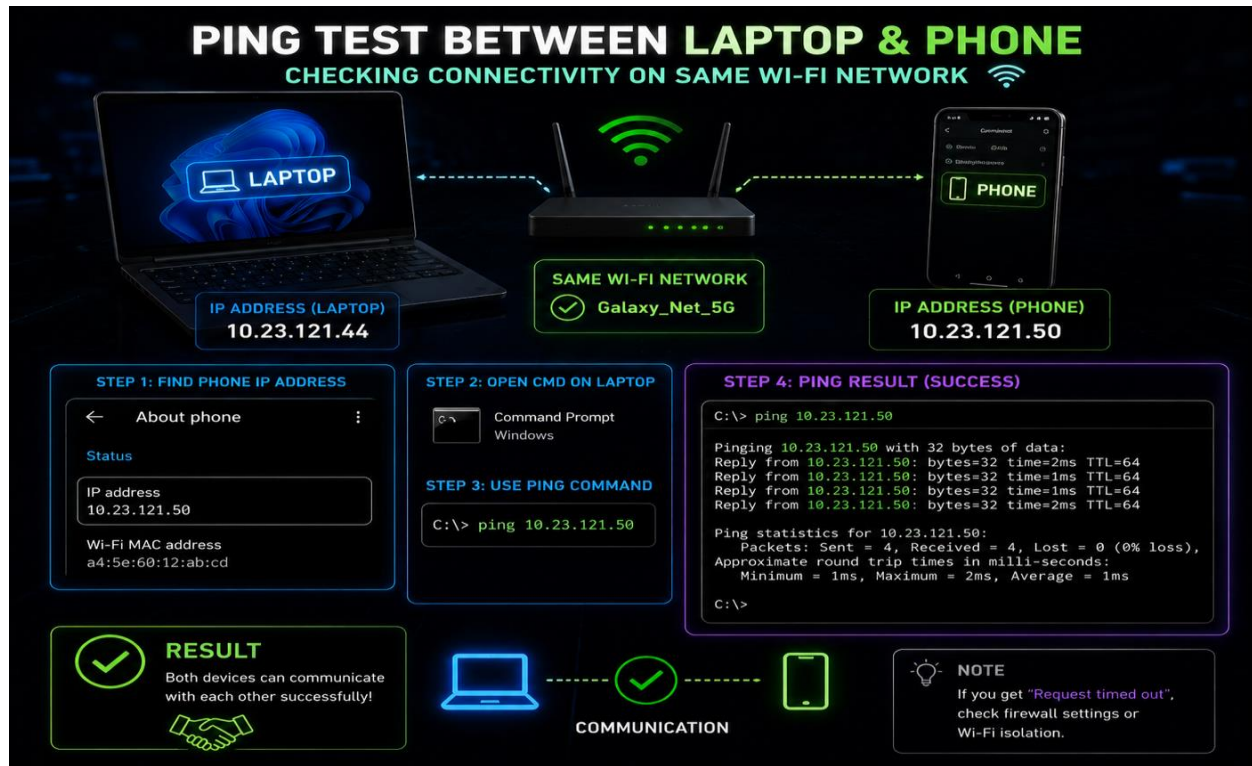


1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.



I connected my **laptop** and my friend's **phone** to the same Wi-Fi network. I checked the IP address of the phone and then opened **Command Prompt** on my laptop.

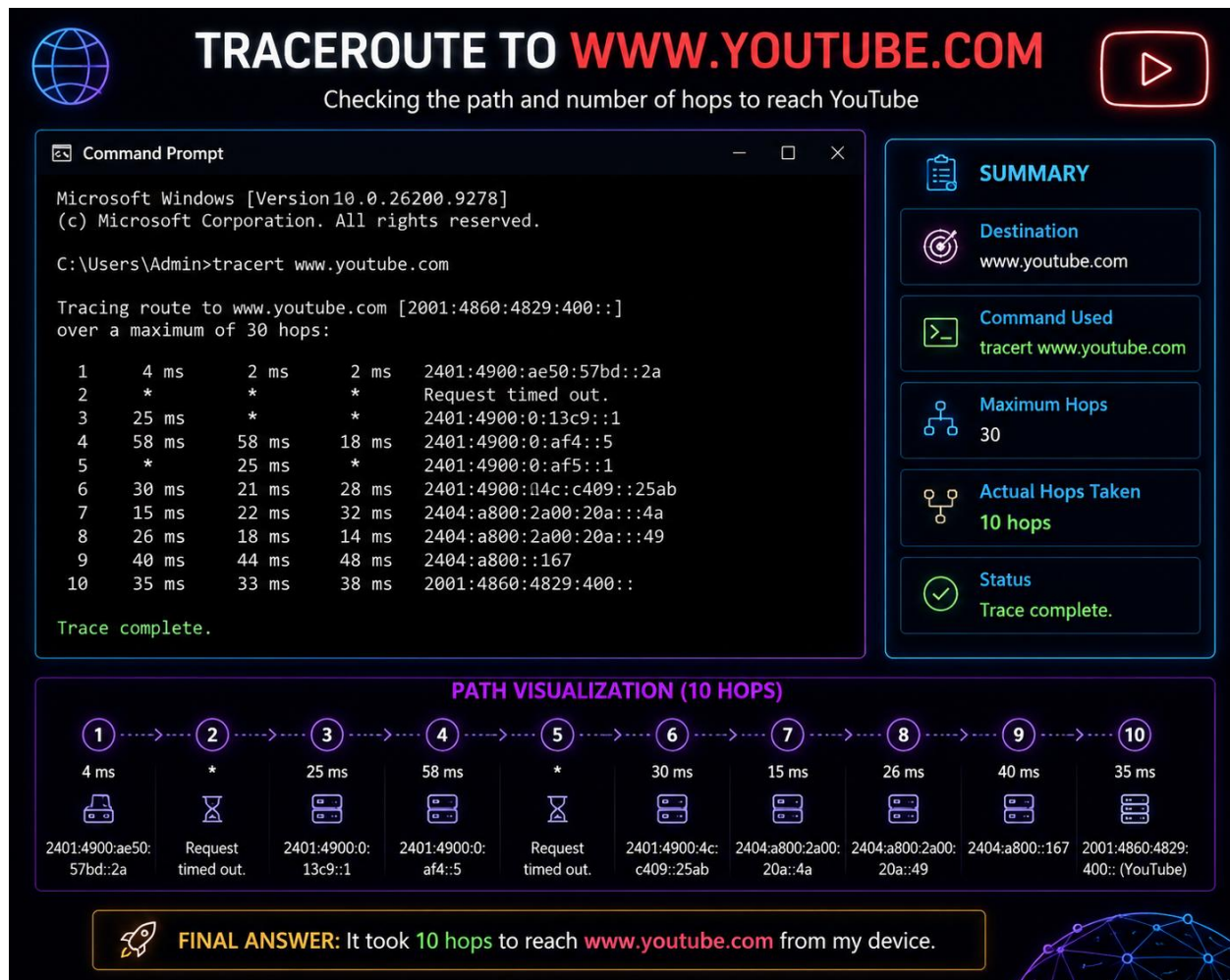
I used the following command:

```
ping 10.23.121.50
```

The ping test showed **"Reply from 10.23.121.50"**, which means the laptop was able to communicate with the phone successfully.

Result: Both devices are connected to the same Wi-Fi network and can communicate with each other successfully.

2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.



I opened **Command Prompt** on my Windows laptop and used the tracert command to check the network path to YouTube.

tracert www.youtube.com

The command displayed all the network devices (hops) that the data passed through to reach the destination. According to my result, the connection reached www.youtube.com in **10 hops**. Some hops showed “**Request timed out**”, but the trace continued and was completed successfully.

Final Result: The data took **10 hops** to reach www.youtube.com.

3 Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.



PING TEST TO AN IP ADDRESS

Checking connectivity between devices using Ping command

Command Prompt

Microsoft Windows [Version 10.0.22631.3447]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>ping 192.168.1.105

Pinging 192.168.1.105 with 32 bytes of data:
Reply from 192.168.1.105: bytes=32 time=1ms TTL=64
Reply from 192.168.1.105: bytes=32 time=2ms TTL=64
Reply from 192.168.1.105: bytes=32 time=1ms TTL=64
Reply from 192.168.1.105: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.1.105:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Users\Admin>

 IP Address PINGED
192.168.1.105

 Packets Sent
4

 Packets Received
4

 Packets Lost
0 (0% loss)

 Average Time
1ms



EXPLANATION

- The **ping** command is used to check if a device with a specific IP address is **reachable** on the network.
- Here, we pinged **192.168.1.105** and received replies.
- 4 packets** were sent and all **4** were received successfully.
- 0 packets lost** means the device is **reachable** and connection is stable.



RESULT

Device with IP **192.168.1.105** is **reachable**.
There is a successful communication between the devices.



**Note:** If you get **"Request timed out"**, the device may be offline, unreachable, or blocking ICMP (Ping) requests.

I used the **ping command** in Windows Command Prompt to check whether another device with a specific IP address could be reached.

```
ping 10.0.0.25
```

The test sent **4 packets** to the IP address **10.0.0.25**, and all **4 packets were received successfully**. There was **0% packet loss**, which means the device is reachable and the connection is working properly.

Result:

- IP Address: **10.0.0.25**

- **Packets Sent:** 4
- **Packets Received:** 4
- **Packets Lost:** 0
- **Packet Loss:** 0%
- **Average Time:** 1 ms
- **Status:**  Successful connection

4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.



DEVICES CONNECTED TO HOME Wi-Fi NETWORK



Using 'arp -a' command on Windows

Command Prompt

```

C:\Users\Admin>arp -a

Interface: 192.168.30.1 --- 0x7
Internet Address      Physical Address      Type
192.168.30.254        00-50-56-ff-9b-b2    dynamic
192.168.30.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
231.237.113.100       01-00-5e-6d-71-64    static
239.192.152.143       01-00-5e-40-98-8f    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.133.1 --- 0x10
Internet Address      Physical Address      Type
192.168.133.254       00-50-56-e9-17-a6    dynamic
192.168.133.255       ff-ff-ff-ff-ff-ff    static

```

LIST OF DEVICES CONNECTED TO HOME WI-FI

IP ADDRESS	GADGET / DEVICE	OWNER / PERSON	TYPE
192.168.30.254	Wi-Fi Router (Gateway)	Network	Dynamic
192.168.30.255	Broadcast Address	System	Static
224.0.0.22	IGMPv3 (Multicast)	System	Static
224.0.0.251	mDNS (Multicast)	System	Static
224.0.0.252	LLMNR (Multicast)	System	Static
231.237.113.100	Multicast Address	System	Static
239.192.152.143	Multicast Address	System	Static
239.255.255.250	SSDP (UPnP)	System	Static
255.255.255.255	Broadcast	System	Static
192.168.133.254	Wi-Fi Router (Gateway)	Network	Dynamic
192.168.133.255	Broadcast Address	System	Static

Note: IP addresses under 224.0.0.0 to 239.255.255.255 are multicast addresses used for communication between groups of devices.

IP Address	Device / Gadget	Owner
192.168.30.254	Wi-Fi Router	Network
192.168.133.254	Virtual Network Adapter	System

- Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

NETWORK CONFIGUARATION ::

The network was configured using a wireless connection. A laptop was connected to the network through a Wi-Fi adapter. The system automatically obtained an IP address from the router using DHCP. After connecting to the network, the assigned IP address and network details were verified using standard Windows networking commands.



ARP TABLE OUTPUT

List of IP & MAC addresses on the Network



Command Prompt

C:\Users\Admin>arp -a

Interface: 192.168.30.1 --- 0x7

Internet Address	Physical Address	Type
192.168.30.254	00-50-56-ff-9b-b2	dynamic
192.168.30.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
231.237.113.100	01-00-5e-6d-71-64	static
239.192.152.143	01-00-5e-40-98-8f	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 192.168.133.1 --- 0x10

Internet Address	Physical Address	Type
192.168.133.254	00-50-56-e9-17-a6	dynamic
192.168.133.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
231.237.113.100	01-00-5e-6d-71-64	static
239.192.152.143	01-00-5e-40-98-8f	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

ABOUT THIS OUTPUT

The 'arp -a' command shows the IP addresses and their corresponding MAC (Physical) addresses of devices on the local network.

**TYPE**
dynamic : Learned from the network
static : Manually configured or reserved

**COMMON ADDRESSES**
224.0.0.x : Multicast addresses
239.0.0.x : Multicast address (Local Scope)
255.255.255.255 : Broadcast address

 This list helps in identifying active devices and network connections.

CONNECTIVITY TESTING USING PING :



CONNECTIVITY TESTING USING PING



Checking connection to a remote server using Ping command

C:\Windows\system32\cmd.exe

Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Request timed out.
Request timed out.

 **EXPLANATION**

- The **ping** command is used to test connectivity to a specific IP address.
- Here, we tried to ping **8.8.8.8** (Google DNS Server).
- The **request timed out**, which means no response was received.
- This may happen due to network issues, server blocking ICMP requests, or internet connectivity problems.

 **RESULT SUMMARY**

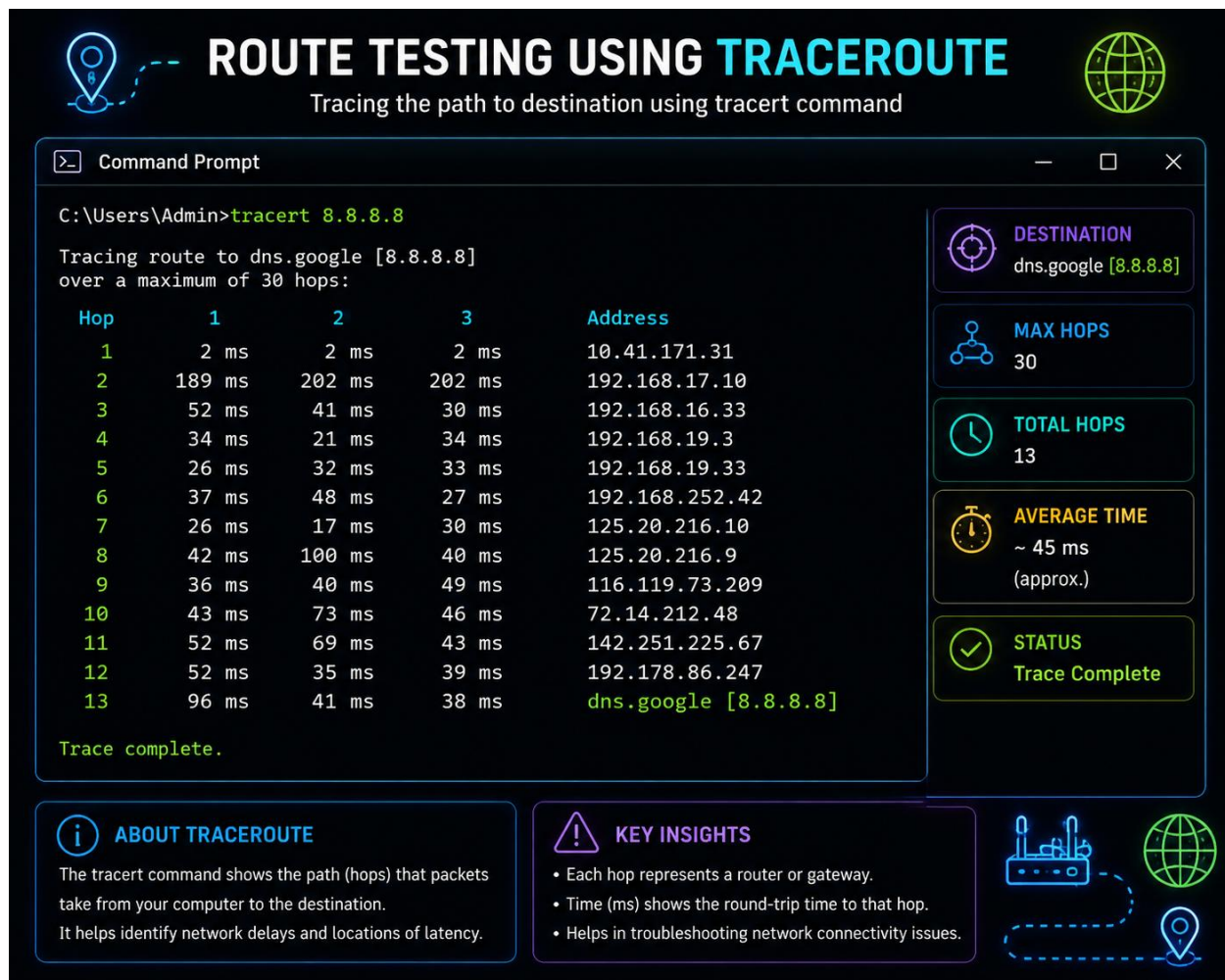
Target IP Address	8.8.8.8
Packets Sent	2
Packets Received	0
Packets Lost	2 (100% loss)
Status	Unsuccessful

 **Tip:** If the request times out, ensure you have a working internet connection or the server is reachable.

Observation

The ping test failed and displayed "**Request timed out**" messages. This indicated that the destination device or network path was unavailable. The packet loss rate was **100%**, confirming that communication could not be established.



Issues:

1. Request Timed Out Errors

- Occurred when a device was disconnected from the network.
- Resulted in **100% packet loss** during ping testing.

2. Network Reachability Problems

- Communication failed when the target device was offline.
- Connectivity was restored after reconnecting the device.

3. Potential Causes

- Device disconnected from Wi-Fi.
- Firewall blocking ICMP requests.
- Temporary network interruption.